

Alexey Shvechkov

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Summary

Principal Software Engineer with 20+ years architecting resilient systems—from Linux kernel drivers to cloud-native AI platforms. Recent focus: ML-powered ransomware detection (70% faster recovery), agentic RAG chatbot (15% support reduction), and Kubernetes-based Azure protection SaaS. Deep expertise in C++, Go, Python, PyTorch. Led distributed engineering teams while prototyping and delivering production systems at enterprise scale. Patent holder and open-source contributor.

Core Competencies /Skills

Leadership & Strategy: Technical strategy, agile execution, product ownership, cross-functional influence, managing & mentoring teams (10+ years), roadmap planning, cost optimization

Cloud & Distributed Systems: AWS (EC2, S3, Lambda, API Gateway), Kubernetes, Docker, Terraform, scalable microservices, high-availability architectures, CI/CD pipelines, fault tolerance

Programming Languages: C++, Go, Python, Rust, Perl, Bash

AI/ML: Generative AI, RAG, transformers (LoRA fine-tuning), PyTorch, TensorFlow, Hugging Face, LangChain, ONNX Runtime, distributed training, anomaly detection

Systems Programming: Linux/Windows kernel drivers (eBPF, KMDF/WDM, minifilters), file systems, network programming, performance tuning

Databases & Storage: PostgreSQL, MySQL, Redis, Cassandra, object storage systems

Frontend & APIs: React, Next.js, REST API design, rapid prototyping

Professional Experience

Arcserve — Technical Director, Software Engineering | 2014 – Present — Greater Boston, MA

*** highlights / most recent projects from last 3 years ***

Anomaly & Ransomware Detection for Backup/Storage

- Architected and implemented engine for detecting anomalies and traces of ransomware activities in air gapped backups/storage using ML/statistical models; Designed and trained various ML/AI models (including BERT based LLMs) for threat classifications/detection; Built automated training and telemetry processing pipelines for models' evaluation and quality assessment. Technically led and managed team of developers ("innovation group"); According to initial studies this feature reduced ransomware incident response & recovery times by ~70%
- Tech stack: Windows internals (MFT, ReFS, USN parsing, mini-filters), ML/AI (Random Forest, XGBoost, feature engineering), BERT based LLMs (training & fine-tuning), FFT + Isolation Forest, Prophet, FAISS, ONNX Runtime, YARA, AWS Glue/SageMaker, C/C++, Python (NumPy, TensorFlow, Hugging Face).

Agentic RAG Chatbot for Troubleshooting & Monitoring SaaS and on-Prem products

- Designed and implemented (end-to-end) multimodal chatbot integrated with products via APIs/MCP for products' monitoring, troubleshooting, and documentation-based Q&A. Also, bot provided UI visual guidance, detected anomalies, performed tasks, reduced support calls by ~15%.
- Tech stack: OpenAI API, LangChain, RAG, Ollama, Hugging Face, AWS Lambda (Python), Python FastAPI & FastMCP, React, HTML/CSS.

Azure Workloads Protection (Backup & Replication)

- Architected and prototyped a massively parallel/scalable Kubernetes-based SaaS solution to protect and restore Azure workloads (VMs, containers, k8s, blob storage); Implemented critical parts of the system; Led a team of developers throughout design and implementation phases.
- Tech stack: Golang, Azure APIs/Golang SDK, k8s, containers, Terraform, gRPC/Protobuf, Redis (KV store, pub/sub).

Cyber-Resilient Storage Appliance

- Architected and prototyped an immutable storage appliance with hardened Linux + OpenZFS, supporting self-healing storage, immutable snapshots, and backup integration; Accelerated product delivery by 30–40% via reusable prototypes and code contributions.
- Tech stack: Linux internals, OpenZFS, Python (backend/CLI), Golang (S3-compatible connector).

Ransomware Protection Module for Windows Storage product

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- Developed Windows minifilter driver and user-mode service to protect backup product services/data from ransomware tampering (Windows equivalent of AppArmor).

- Tech stack: Windows internals, minifilters (IFS/WDK).

Additional Contributions

- Managed small/medium sized distributed teams (5-15) developing & maintaining following products:

- High availability & real-time full host/VM replication product (aka RHA).

- Highly scalable archival storage appliance with distributed fault-tolerant FS (aka OneXafe).

- SaaS-based backup of on-prem data (aka Cloud Direct).

- Contributed as IC fixing bugs in legacy products and as domain expert/architect on new feature development.

- Actively participated in hackathons (winning awards); Prototyped innovative product features; Technically lead and manage team responsible for company's AI/ML strategy ("innovation group").

- Mentoring, Hiring, Managing teams, performance reviews

CA Technologies (formerly Computer Associates) — Principal Engineer | 2006–2014 — Greater Boston, MA

- Led post-acquisition integration of XOssoft's replication technologies into CA's enterprise portfolio; Designed and developed high-availability replication, deduplication, and backup features using C/C++ for Windows and Linux; Built a Unix/Linux build automation framework with web UI, reducing manual labor by 50%.

XOssoft (Acquired by CA Technologies in 2006) — Lead Software Engineer | 2000–2006 — Israel

- Designed and developed CDN components, including network management and monitoring tools, and high-traffic HTTP/FTP caching proxies, along with replication solutions; Led the development of kernel-mode drivers and file system modules in C/C++, improving system performance and reliability.

Yandex — Moscow, Russia Software Engineer | 1998–2000

- Developed high-traffic search, advertising network and web applications using C++ and Perl, optimizing performance for large-scale systems.

Open-Source Projects/POCs (examples)

- [s3stor](#) – deduplicating archiving /backups into S3 compatible storage (Go) - github.com/shvechkov/s3stor

- [gos3rve](#) – exposing local file systems via s3 APIs (Go) - github.com/shvechkov/gos3rve

- [ufc](#) – fast unique file copy/ indexes and stores unique files (Rust) - github.com/shvechkov/ufc

Education

- M.Sc., Applied Mathematics | Institute of Telecommunications and Computer Science, Russia | 1996

- Machine Learning Specialization (Stanford Online) | 2024 ([Certificate](#))

- Holder of multiple [U.S. patents](#) in storage and infrastructure technologies



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